

myCompiler

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FILES

main.m

```
1 clear;
2 close all;
3 clc;
4
5 %% Parameters
6 Fs = 600e3; % Sampling frequency
7 t = 0:1/Fs:5e-3; % 5 ms duration
8
9 fm = 5e3; % Message frequency
10 fc = 20e3; % Carrier frequency
11
12 %% Message Signal
13 m = sin(2*pi*fm*t);
14
15 %% Carrier Signal
16 c = cos(2*pi*fc*t);
17
18 %% DSB-SC Modulated Signal
19 s = m .* c;
20
21 %% ----- PLOTS -----
22
23 figure;
24
25 subplot(4,1,1)
26 plot(t(1:1000),m(1:1000),'LineWidth',1.3)
27 title('Message Signal (5 kHz)')
28 xlabel('Time (s)')
29 ylabel('Amplitude')
30 grid on
31
32 subplot(4,1,2)
33 plot(t(1:1000),c(1:1000),'LineWidth',1.3)
34 title('Carrier Signal (20 kHz)')
35 xlabel('Time (s)')
36 ylabel('Amplitude')
37 grid on
```

Run Save Output Input

Detected Sideband Frequencies:
Peak 1 = 15.0 kHz
Peak 2 = 25.0 kHz

[Execution complete with exit code 0]

Blocks.ai

Reach your agent anywhere. No inbound ports, no DNS, no static IP, any device, behind NAT. Blocks.ai →

Ads by EthicalAds

Message Signal (5 kHz)

Center Signal (20 kHz)

DSB-SC Modulated Signal

Spectrum of DSB-SC Signal

